Power System Protection

IEEE Press
445 Hoes Lane
Piscataway, NJ 08854

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Power System Protection

Second Edition

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IEEE Press Series on Power and Energy Systems
Ganesh Kumar Venayagamoorthy, Series Editor

IEEE PRESS
WILEY

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.
Published simultaneously in Canada.

Library of Congress Cataloging-in-Publication Data applied for:
ISBN: 9781119513148

Cover design by Wiley
Cover image: © INTERMOUNTAIN AC SWITCHYARD BY A.P “SAKIS” MELIOPOULOS

Set in 9.5/12.5pt STIXTwoText by Straive, Chennai, India

10 9 8 7 6 5 4 3 2 1

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Author Biographies

Charles Henville has more than 50 years of experience as a power system engineer, including 40 years as a protection engineer. He worked for 30 years as a commissioning engineer and protection engineer for a large Canadian utility before starting his own consulting engineering company. He has been an active participant in the IEEE Power and Energy Society, Power System Relaying and Control Committee (PSRCC). This committee is responsible for creating numerous IEEE standards and technical papers dealing with power system protection. Charles was the Chairman of that Committee and one of its subcommittees and of several of its working groups.

He is also actively involved in teaching and training working engineers. He has presented several short courses for continuing professional development and has served as an instructor at Gonzaga University, the University of Wisconsin, and the University of British Columbia. As a trainer for utility engineers, he has had broad exposure to protection practices in North America and worldwide.

Rasheek Rifaat (P.Eng., IEEE Life Fellow) graduated from Cairo University (Egypt), 1972, BSc and McGill University, (Montreal, QC, Canada), 1979, Master Engineering. He has over 45 years of experience with industrial and utility electrical power systems in Canada, North America, and overseas. His experience extends to renewable and traditional power generation, cogeneration, industrial plants, power transmission, sub-transmission and distribution systems, and large system interconnections. His long time experience encompasses power system protection design, coordination and settings, and power system studies including transient studies, insulation coordination, surge protection, power system reliability, and operational requirements.

He authored and presented over 40 peer-reviewed technical papers and publications and presented many tutorials on power system protection and transient studies. He is also the current Chair of the Work Group looking after the IEEE Standards for Protection and Coordination of Industrial and Commercial Power Systems (formerly known as the Buff Book, Reproduced as Standards Series 3004). He has been involved with the Institute of Electrical and Electronic Engineers Inc. (IEEE) for over 40 years and became a Life Fellow Member since 2016. He has been awarded the year 2020, R. H. Tanner Award for industry leadership for the Canadian Region (Region 7).

He is presently registered as a P.Eng. in Alberta, Saskatchewan, and Ontario and is a member of the Energy Industry Electrical Engineering Association.

Brian Johnson is a University Distinguished Professor and the Schweitzer Engineering Laboratories Endowed Chair in Power Engineering in the University of Idaho Department of Electrical and Computer Engineering. He received BS, MS, and PhD degrees in electrical engineering from the University of Wisconsin-Madison. He has worked at the University of Idaho for over 25 years, where he was chair of the Department of Electrical and Computer Engineering from 2006 to 2012.

His teaching and research interests include power system protection, HVDC transmission, power electronic applications in power systems, and resilience controls for critical infrastructure systems. He is active with the IEEE Power and Energy Society, where he was the chair of the Power and Energy Education Committee from 2014 to 2015 and of the IEEE HVDC and FACTS subcommittee in 2018–2020. He is currently a member of the editorial board for IEEE Power and Energy Magazine. He is a registered professional engineer in the State of Idaho.

Sakis Meliopoulos obtained a Diploma in Electrical and Mechanical Engineering from the National Technical University in Athens, Greece, in 1972 and a Master in EE (1974) and a PhD degree (1976) from the Georgia Institute of Technology in Atlanta, Georgia, USA. He joined the faculty of the Georgia Institute of Technology as an Assistant Professor (1976), Associate Professor (1982–1988) and full professor (1989 to present). In 2006 he was named the Georgia Power Distinguished Professor. He is actively involved in education and research for improved safety and electromagnetic compatibility of electric power installations, protection, and control of power systems and the application of new technology in these areas. Since 1999 he is the Georgia Tech Site Director of PSERC, an NSF I/URC. Since 2016 he is the associate director for cyber-physical security of the Georgia Tech Institute for Information Security and Privacy (IISP). He has pioneered several new analysis and design techniques for bulk power reliability analysis, safety, protection, and electromagnetic compatibility of electric power systems. Most well-known is the EPRI transmission reliability program TRELLS (now renamed TransCARE), the GPS-synchronized harmonic state measurement system for transmission systems (first [1993] wide area measurement system on NYPA), the distributed dynamic state estimation method (SuperCalibrator), the setting-less relay, the CYMSA software (Cyber-Physical Modeling and Simulation for Situational Awareness), his invention of the Smart Ground Multimeter, the EPRI grounding analysis programs, and the WinIGS (Integrated Grounding System analysis and design). He has modernized many power system courses at Georgia Tech, initiated the power system certificate program for practicing engineers, and he co-developed the Master of Science Cybersecurity, Energy Systems degree. He is a Fellow of the IEEE. He holds three patents, and he has published three books, and over 430 technical papers. He has received a number of awards, including the Sigma Xi Young Faculty award (1981), the outstanding Continuing Education Award, Georgia Institute of Technology (twice 2002 and 2014), the 2017 D. Scott Wills ECE Distinguished Mentor Award, the 2020 ECE Distinguished Faculty Achievement Award, he received the 2005 IEEE Richard Kaufman Award and the 2010 George Montefiore international award. In 2019 he received the Doctor Honoris Causa from his alma matter, NTUA.

Preface to the Second Edition

In the First Edition, Paul Anderson built a solid foundation of analytical study of electric power system protection. Most of the analytical techniques are still applicable. During the 23 years since the first publication, there have been many advances in the technology that are now incorporated in the second publication. Such advances included two fronts. The first front has been associated with the gradual reformatting of the overall system with more renewable generation, high efficiency components, and increased interests in safe and secure operations of the power system. The second front has been associated with the applications of numeric multifunction relays with further intelligence and abilities in recognizing fault occurrence, and in developing protection operational decisions. With the continuation of such advances, protection engineers and specialist need a solid understanding of the correlated concepts discussed in the book.

The first edition was a massive work for a single author. This second edition needed the combined efforts of four the authors to bring the book up to date.

Most of the review work was provided by internal review between various authors other than the initial author/updater. However, we gratefully acknowledge additional thoughtful and helpful reviews by Vijay Vittal and Ralph Barone.

In addition to updating references and standards and correcting a few typographical errors, the second edition now includes the following:

In Part 1, *Protective Devices and Controls*, issues such as power quality and regulatory requirements are updated. New technologies such as phasor measurements and proliferation of precise time applications are noted. The impact of IEC standards on fuse ratings and peer-to-peer communications is discussed. Six generations of protective relays are now noted in Chapter 3 instead of the four in the First Edition. These additional generations are discussed in appropriate later chapters. The standardization of digital overcurrent relays is described, together with migration from using graphical time current characteristics to calculated ones. Inverter based resources now have an impact on network faults.

In Part 2, *Protection Concepts*, we add discussion of different measuring principles such as negative sequence overcurrent, new methods of polarizing directional ground overcurrent relays, coordinating overcurrent relays measuring currents at different voltages, and transmission line protection using quadrilateral distance relays. Analytical techniques for distance relays in Chapters 8–10 remain largely unchanged.

In Part 3, *Transmission Protection*, we add discussion of issues that have arisen since the first edition. These include the following:

- A detailed settings example for a three zone quadrilateral distance relay
- Impact of frequency changes on the security of distance relays
- New definitions of short and long lines

- New techniques of line differential protection
- New teleprotection applications
- Detection of secondary arc extinction facilitating single phase tripping and reclosing

In Part 4, *Apparatus Protection*, the scope is expanded to include industrial protection systems. Modern differential protection systems using advance digital relays are discussed. We discuss issues related to reliability of multifunction systems such as “putting all eggs in one basket.” With increased concerns linked to excessive arc flash incident energy in industrial system, bus protection aspects and applications have been considered, and associated descriptions added.

In Part 5, *System Aspects of Protection*, we explore the meaning of system integrity protection schemes (SIPS) as a development following special protection systems (SPS). System stability issues are expanded to include voltage stability concerns. In addition, two new chapters are added to address modern developments. New chapters are

- *Voltage Sourced Converter Protection*, which addresses the application and protection of modular multi-level converters. These converters are commonly used for interconnecting wind generation, photovoltaic sources, and batteries.
- *Protection of Independent Power Producer Interconnections*, which addresses the issues facing interconnection of market energy sources to transmission and distribution systems.

In Part 6, *Reliability of Protective Systems,* there are relatively few changes as the analytical concepts have not changed significantly since the first edition.

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List of Symbols

The general rules used for symbols in this book are as follows. Letter symbols for units of physical quantities (unit symbol abbreviations) are given in Roman or Greek typeface, e.g. A for amperes, VA for voltamperes, or Ω for ohms. Mathematical variables are always given in italic typeface, and may be either Roman or Greek, e.g. v for instantaneous voltage, V for rms phasor voltage magnitude, or λ for failure rate of a component. Complex numbers, such as phasors, are given in bold italic typeface, e.g. $\boldsymbol{I}$ for current and $\boldsymbol{V}$ for voltage. Matrices are specified by bold Roman typeface, e.g. **Z** for an impedance matrix. The following tabulations give specific examples of symbols used in this book.

1. Capitals

A	ampere; unit symbol abbreviation for current
$\boldsymbol{A}$	complex two port network transmission parameter
B	= Im $\boldsymbol{Y}$, susceptance
$\boldsymbol{B}$	complex two-port transmission parameter
C	capacitance
CTI	coordination time interval
$\boldsymbol{C}$	complex two-port transmission parameter
D	damping constant
$\boldsymbol{D}$	complex two-port transmission parameter
E	source emf; voltage
E_S	source voltage, relay equivalent circuit
E_U	source voltage, relay equivalent circuit
F	failure probability
F	farad; unit symbol abbreviation for capacitance
G	= Re $\boldsymbol{Y}$, conductance
H	inertia constant
H	henry, unit symbol abbreviation for inductance
Hz	hertz, unit symbol abbreviation for frequency
I	rms current magnitude
$\boldsymbol{I}$	rms phasor current
K	spring constant; controller gain
L	inductance
LL	line-to-line

LN	line-to-neutral
M	$=10^6$, mega, a prefix
M	matrix defining relay voltage and current
MOC	minimum operating current, overcurrent relay
N	newton, unit symbol abbreviation for force
P	average real power
P_n	natural power of a transmission line, *SIL*
Q	average reactive power, unavailability
R	$=\mathrm{Re}\,\boldsymbol{Z}$, resistance
S	$=P+jQ$, complex apparent power
S_B	base complex or apparent power, VA
SIL	surge impedance loading of a transmission line
T	time constant
V	rms voltage magnitude
$\boldsymbol{V}$	rms phasor voltage
V	volt, unit symbol abbreviation for voltage
VA	volt-ampere, unit symbol abbreviation for apparent power
VAR	volt-ampere, reactive, unit symbol abbreviation for var
W	number of failures in a specified time period
W	watt, unit symbol abbreviation for active power
X	$=\mathrm{Im}\,\boldsymbol{Z}$, reactance
$\boldsymbol{Y}$	$=G+jB$, complex admittance
$\boldsymbol{Y}_C$	characteristic admittance of a transmission line $+1/\boldsymbol{Z}_C$
Y	admittance magnitude
Y	admittance matrix
$\boldsymbol{Z}$	$=R+jX$, complex impedance
$\boldsymbol{Z}_R$	impedance seen at the relay at R
$\boldsymbol{Z}_E$	external system impedance of a network equivalent
$\boldsymbol{Z}_S$	source impedance of a network equivalent
$\boldsymbol{Z}_U$	source impedance of a network equivalent
$\boldsymbol{Z}_F$	fault impedance
Z	impedance magnitude
Z	impedance matrix

2. Lowercase

ac	alternating current
a–b–c	phase designation of three phase currents or voltages
b	$=\omega c$, line susceptance per unit length
c	capacitance per unit length
dc	direct current
det	determinant of a matrix
e	base for natural logarithms
f	frequency
h	distance measurement parameter
i	instantaneous current designation

j	$\sqrt{-1}$, a 90° operator
k	$= 10^3$, kilo, an mks prefix
k	degree of series compensation
l	inductance per unit length
ℓ	length of a transmission line
In, log	natural (base e) logarithm, base 10 logarithm
m	$= 10^{-3}$, milli…, an mks prefix
m	mutual inductance per unit length
pu	per unit
p	instantaneous power; Maxwell's coefficient
r	apparatus resistance
s	distance along a transmission line
t	time
v	instantaneous voltage
var	voltampere reactive, when used as a noun or adjective
w	frequency of failure
x	line reactance per unit length; apparatus reactance
y	line shunt admittance per unit length
z	line impedance per unit length

3. Uppercase Greek

Δ	delta connection; determinant (of a matrix)
Σ	summation notation
Ω	ohm, unit symbol abbreviation for impedance

4. Lowercase Greek

γ	propagation constant of a transmission line
δ	voltage phase angle
ζ	magnitude of impedance
θ	phase angle of voltage or current, angle of impedance
λ	item failure in reliability calculations
μ	$= 10^{-6}$, micro, an mks prefix
π	pi
ρ	resistivity, magnitude of admittance
τ	time constant
ϕ	phase angle of voltage or current, angle of admittance
ω	radian frequency

5. Subscripts

a	phase a
A	phase a

b	phase b
B	phase b
B	base quantity
c	phase c
C	phase c
LN	line-to-neutral
LL	line-to-line
max	maximum
min	minimum
R	receiving end, of a transmission line
S	sending end, of a transmission line
u	per unit, sometimes used for clarity

6. Superscripts and Overbars

$(\ )^t$	transpose (of a matrix)
$(\ )^{-1}$	inverse (of a matrix)
$\tilde{A}$	tilde, distinguishing mark for various quantities
$\hat{A}$	circumflex, distinguishing mark for various quantities
$(\ )^*$	conjugate, of a phasor or matrix